Experimental Design for Carbon-Aware Orchestrator Publication

1 Comparative Evaluation Framework

Algorithms to Compare:

1. Carbon-Aware Orchestrator
2. Kubernetes Default Scheduler
3. Random Placement
4. First-Fit/Bin-Packing Approach
    - Process workloads in a predefined order (typically arrival time). For each workload, scan through nodes sequentially. Place the workload on the first node with sufficient resources. If no node has capacity, reject the workload or wait.
    - Differences from CAO:
        - Only considers where to place workloads, not when
        - Optimizes for bin utilization, not carbon emissions
        - Makes greedy, immediate decisions without considering future timeslots
        - Has no concept of deadlines or scheduling windows
5. Theoretical optimal solution (if computationally feasible)
6. Other state-of-art energy-aware schedulers

2 Workload Diversity

Synthetic Workloads

Parameterized by characteristics:

– CPU/Memory intensity ratios

– Execution duration (short vs. long-running)

– Deadlines (strict vs. flexible)

– Burstiness patterns

Real-World Traces (tbd)

– Alibaba or Google cluster traces

– CI/CD pipeline workloads

– Batch processing workloads

– ML training jobs

3 Key Metrics

Primary Carbon Metrics

– Total carbon emissions (kg CO_2e)

– Carbon reduction percentage vs. baselines

– Carbon efficiency (computation/kg CO_2e)

System Performance

– Resource utilization across nodes/regions

– Job completion times

– SLA violations/deadline misses

– Scheduling overhead (time complexity)

Tradeoff Analysis

– Carbon savings vs. performance impact

– Pareto efficiency plots

4 Experimental Variables

Systematically vary:

– **Infrastructure Scale:** 4, 8, 16, 32, 64 nodes

– **Regional Distribution:** Single-region vs. multi-region

– **Carbon Intensity Patterns:** Static, time-varying, region-varying

– **Scheduling Window Flexibility:** 0h, 24h, 48h, 72h

5 Statistical Validity

- Run each experiment configuration 30+ times
- Report mean, median, standard deviation, and 95% confidence intervals
- Conduct statistical significance tests (e.g., t-tests, ANOVA)
- Perform sensitivity analysis for key algorithm parameters

6 Practical Impact

- Demonstrate real-world applicability through case studies
- Show financial cost implications (carbon pricing)
- Calculate environmental impact at scale (projections)